

Small choroidal melanoma: outcomes following apical height dose brachytherapy

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Received 27 May 2020

Revised 20 July 2020

Accepted 21 July 2020

Published Online First

2 September 2020

ABSTRACT

Aim To assess the outcomes of small choroidal melanoma following iodine-125 episcleral brachytherapy (apical height dose of 85 Gy).

Methods Patients with small choroidal melanoma that underwent iodine-125 episcleral brachytherapy between January 2004 and December 2017 were reviewed. Inclusion criterion for this study was the COMS small tumour size (tumour apical height of 1.0–2.5 mm and largest basal diameter (LBD) <16.0 mm). Patients that received any form of prior therapy or adjuvant transpupillary thermotherapy were excluded. Outcome measures were visual acuity (VA), recurrence, ocular survival and metastasis at 3 years. Kaplan-Meier estimation was calculated for VA, recurrence, ocular survival and survival outcome (overall and metastasis-free survival rate) at 3 years.

Results 161 cases of choroidal melanoma were included in this study, with the mean (SD) age of 59.6 (14.1) years, and 93 (58%) were males. The mean (SD) apical height for the tumours were 2.1 (0.4) mm and mean (SD) LBD was 8.3 (2.2) mm. The mean (SD, median) follow-up was 40.7 months (37.1, 25 months). The VA was 20/50 or better in 69%. Only one recurrence event (1%) and one enucleation event (1%) were observed. Overall survival was 97%, and no metastatic events were observed at 3 years.

Conclusion Small choroidal melanomas treated with iodine-125 episcleral brachytherapy have excellent outcomes. The majority (69%) of patients retained VA of 20/50 or better with very high local control and ocular survival rate (99.3%) with the absence of metastasis (100%).

INTRODUCTION

There has been a major shift in the management of uveal melanoma away from enucleation since the inception of Collaborative Ocular Melanoma Study (COMS).^{1–3} Most recent Surveillance Epidemiology End Result (SEER) data had revealed that about 70% of all uveal melanoma in the USA are managed by radiation therapy.⁴ Amidst ongoing debate regarding treatment of small choroidal melanoma,^{5–8} there is a trend towards treatment of small choroidal melanoma rather than initial observation to document growth.⁹ In a survey of participants of the International Society of Ocular Oncology, when presented with a case of small choroidal melanocytic lesion, the majority (>60%) favoured treatment with episcleral plaque brachytherapy.¹⁰

Yet, there are significant differences in the application of episcleral plaque brachytherapy for

choroidal melanoma such as choice of radionuclide (iodine-125, palladium-103, ruthenium-106) and the prescription point (scleral base, 5.0 mm or apical height), used for dose calculation.^{11–13} COMS protocol prescribed 85 Gy to the tumour apex with an important exception: for tumours 1<5.0 mm in height, the prescription height was still 5.0 mm (prescription height dose).^{14–15} In contrast, American Brachytherapy Society (ABS) guidelines for tumours that are <5 mm are dosed to their apex and not the 5.0 mm (apical height dose).¹⁶ Although the empiric therapeutic dose of 85 Gy is delivered to the entire tumour with either approach,¹⁷ with COMS protocol, tumours that are <5.0 mm in height receive supratherapeutic dose with greater radiation exposure to adjacent structures. The advantage of ABS guidelines as compared to COMS protocol is of reduced radiation exposure to visually critical structures (optic disc and macula) with potential improved visual outcomes with full therapeutic dose delivered to the tumour.¹⁸

The COMS has not yet published treatment outcomes (vision and tumour control) using its prescription height dose protocol for treatment of small choroidal melanoma with iodine 1–125 episcleral plaque brachytherapy.¹⁹ The limited data regarding outcomes of small choroidal melanoma can be extracted from the published studies (table 1). However, the presence of additional variables such as use of alternative radionuclides (ruthenium-106^{20–22} palladium¹¹ or mixture²³) limits direct comparison. Some studies have included patients that were previously treated or received adjuvant transpupillary thermotherapy (TTT).²³ So as to make our results comparable to future COMS reports, we present outcomes (visual acuity (VA), recurrence, ocular survival and metastasis) in a consecutive cohort of patients diagnosed with small choroidal melanoma (COMS criteria) treated only with iodine-125 episcleral plaque using apical height dose calculations (ABS guidelines).

METHODS

Patients treated for choroidal melanoma between January 2004 and December 2017 were included in the initial clinical chart institutional review board (IRB)-approved review. The inclusion criterion for this study was small tumours per COMS criteria (apical height of 1.0–2.5 mm and largest basal diameter (LBD) <16.0 mm). Patients that received any form of prior therapy or adjuvant TTT were excluded. An exempt status from the IRB at the Cleveland Clinic Foundation was obtained in order to collect and analyse anonymous patient information.



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To cite: Yupari RJ, Bena J, Wilkinson A, et al. *Br J Ophthalmol* 2021;**105**:1161–1165.

Table 1 Small choroidal melanoma: published studies: extracted data

Authors	Year	Isotope	Small tumour (N)	Criteria for diagnosis		Inclusion criteria (apical height)
				Clinical	Histological	
COMS ^{19*}	1997	I-125	188	+	–	COMS
Finger ²⁴	1999	Pd-103	12	+	–	COMS
Damato ²⁰	2005	Ru-106	124	+	–	COMS
Isager ²¹	2006	Ru-106	30	+	–	<6 mm
Verschueren ²²	2010	Ru-106	84	+	–	<3 mm
Sanchez-Tabernero ²⁵	2017	I-125	28	+	–	<2 mm
Shields ^{23†}	2019	I-125, others‡	787	+	–	<2.5 mm
Present study	2020	I-125	161	+	–	COMS

*Visual outcomes not yet reported.

†787 tumours were <2.5 mm to match the COMS criteria.

‡Variety of radionuclides in addition to iodine-125.

§Also includes cases with prior therapy and additional transpupillary thermotherapy.

COMS, Collaborative Ocular Melanoma Study; small choroidal melanoma (apical height of 1–2.5 mm and base 5.0–16.0 mm); I, iodine; Pd, palladium; Ru, ruthenium.

Patient basic demographics and their clinical information were obtained from the electronic medical record system (EpicCare EMR, Epic Systems Corp.) and paper medical records. A plaque simulator treatment planning software (Plaque Simulator V.5.3.4; Bebig GmbH, Berlin, Germany) was used for determining dosimetry for the radioactive plaques. We used the dose–volume histogram to assist in choosing the best plaque (COMS 10–COMS 20, COMS notched and Eye Physics 917). Iodine-125 was the isotope used in the study with apical dose prescription of 85 Gy. The patients received a comprehensive initial workup at the time of presentation. They underwent a detailed ophthalmoscopic examination, photography, and ultrasonographic A and B scans. Tumour size (height and basal diameter) was determined by ultrasound and ophthalmoscopic examination. Accurate plaque placement before the introduction of intraoperative ultrasonographic confirmation in June 2006 was assessed by use of indirect ophthalmoscope or transillumination.²⁶ Patients were followed using a standard clinical protocol of eye examinations every 3 months for the first year, every 6 months for up to 5 years and annually thereafter. Metastasis was excluded at baseline by systemic staging (CT scan of chest abdomen and pelvis) followed by surveillance using hepatic ultrasonography every 6 months for the first 5 years and annually, thereafter.²⁷

Best-corrected visual acuity was recorded using a Snellen eye chart measured at every visit and was converted into logarithm of the minimum angle of resolution values for statistical analysis. Kaplan-Meier estimation was performed to show time to VA of 20/50 or better and to VA of 20/200 or better.

Local recurrence was determined by tumour growth of >0.2 mm in apical height and >0.5 mm in basal diameter compared to pre-treatment tumour dimensions.²⁸ Kaplan-Meier estimation was performed to show time to recurrence. Ocular survival was calculated as the absence of enucleation. Survival outcomes were reported as overall survival (OS) and metastasis-free survival (MFS) rate. Kaplan-Meier estimation was performed to show time to death and death due to metastatic events.

Table 2 Small choroidal melanoma: study demographics (n=161)

Factor	Mean±SD
	Value
Eye	
R	84 (52.2)
L	77 (47.8)
Age (years)	59.6±14.1
Sex	
Female	68 (42.2)
Male	93 (57.8)
Follow-up (months)	40.7 (37.1)
Height (mm)	2.1±0.4
Largest basal diameter (mm)	8.3±2.2
Distance to macula (mm)	4.7±4.9
Distance to optic disc (mm)	5.3±4.1
Prior or adjuvant therapy	0
Total dose to apex (Gy)	85.6±11.0
Status	
Alive	153 (95.0)
Dead due to metastasis	0 (0)
Dead due to other causes	8 (5.2)

L, left; R, right.

RESULTS

A total of 161 patients treated with iodine-125 plaque brachytherapy between 1 January 2004 and 31 December 2017 were included in this study. The mean (SD) age of the included patients was 59.6 (14.1) years at the start of treatment. Ninety-three (58%) were male. The mean (SD) apical height for the tumours were 2.1 (0.4) mm, and mean (SD) large basal diameter was 8.3 (2.2) mm. The mean (SD, median) follow-up was 40.7 months (37.1, 25 months) (table 2).

Best-corrected visual acuity distribution at baseline showed that 74% (119) of patients had a VA of 20/50 or better and 96% (154) had a VA of 20/200 or better. Kaplan-Meier estimation for 3-year event-free rate of VA of 20/50 or better and VA of 20/200 or better was 69% and 87%, respectively (figure 1).

Only one event of local recurrence at 3 years following treatment was observed in the study (confirmed by ultrasonography and fundus photographs), with a 3-year event-free rate of 99.3% (figure 2A). Only one event of enucleation at 3 years (for local recurrence) was observed, with a 3-year event-free rate of 99.3% (figure 2B).

None of the study patients developed metastasis. Eight patients in this cohort died of unrelated causes. OS and MFS were 97% and 100% at 3 years, respectively (figure 2C).

DISCUSSION

To minimise selection bias, we have included consecutive case series using strict criteria to make our results comparable to published studies and future COMS reports. Given the current standard of care, all the included cases were treated based upon clinical diagnosis rather than cytological or histological confirmation. All patients treated at the Department of Ophthalmic Oncology of the Cole Eye Institute were followed using a standardised protocol.²⁸ We report on all clinically relevant outcomes following brachytherapy of choroidal melanoma such as VA, local recurrence, ocular survival and metastasis at 3 years. Such data offer a benchmark against which novel therapies can be assessed.

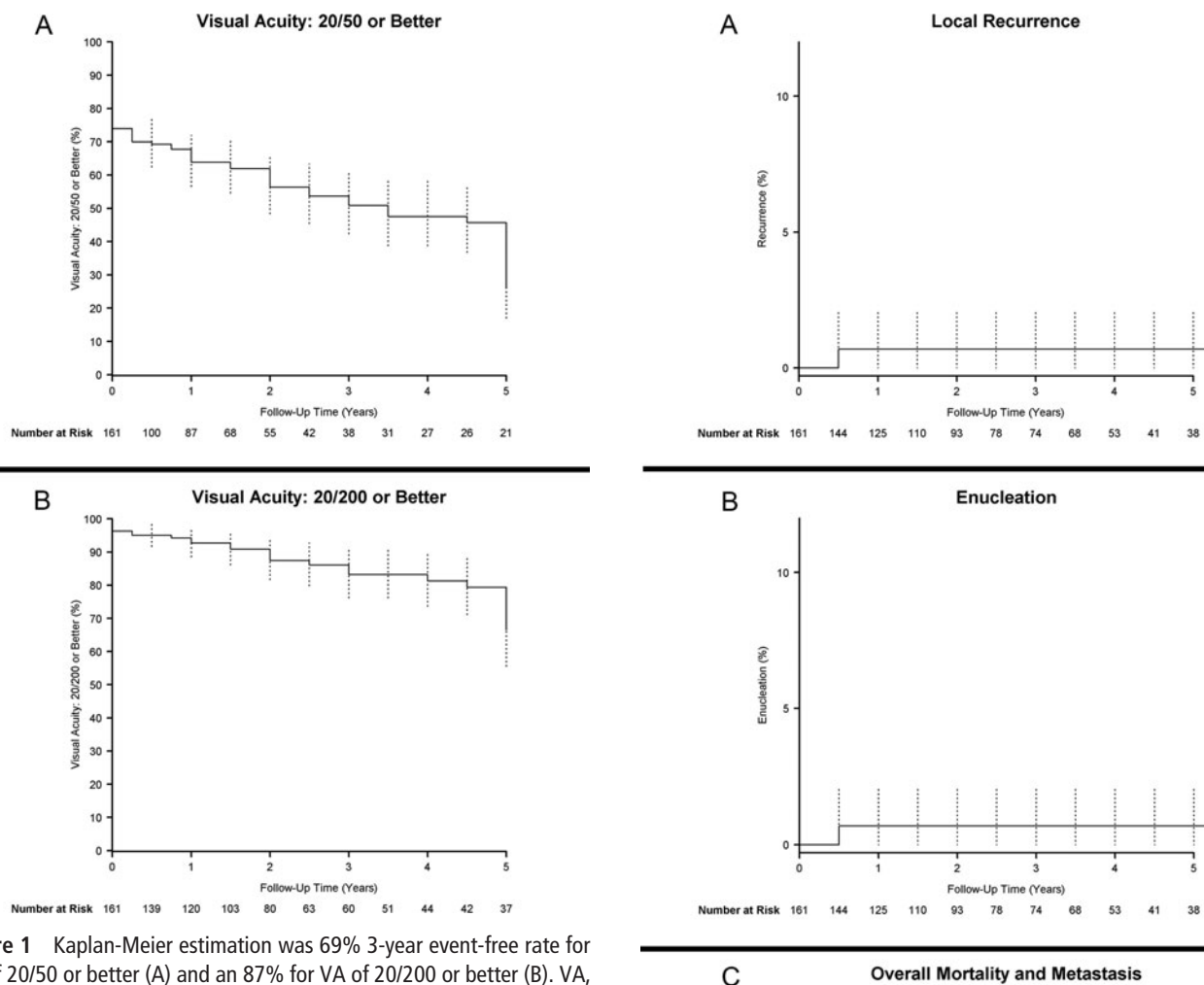


Figure 1 Kaplan-Meier estimation was 69% 3-year event-free rate for VA of 20/50 or better (A) and an 87% for VA of 20/20 or better (B). VA, visual acuity.

Best-corrected visual acuity was recorded at all instances. The VA outcomes (VA 20/50 or better at 3 years) of small choroidal melanoma in our study were comparable with prior studies^{21 22 24} and were better than those reported for medium-sized tumours by the COMS group (58% vs 34%) (figure 1).¹⁵ Although some progress is being made with the use of steroids and anti-vascular endothelial growth factor agents, radiation retinopathy/optic neuropathy are still major complications of brachytherapy.²⁹

All cases were treated using a single radionuclide (iodine-125) with planned apical height dose of 85 Gy as per ABS guidelines.¹⁶ As per COMS protocol, all tumours included in this study would have been treated to the prescription height of 5.0 mm. Since increased tumour size is an important predictor of vision loss,³⁰ it can be assumed that higher dose would have led to higher vision loss, although at present, no such comparative data are available for small tumours. In addition, proximity of the tumour to the disc and macula, by influencing dose exposure to these visually critical structures may also affect the visual outcome.³⁰

We are only aware of one study where comparison of outcomes following brachytherapy using COMS protocol and ABS guidelines (for those patients that were not enrolled in COMS) was performed in non-randomised fashion. Even though some of the treated tumours were of medium size, the authors did not notice any differences in VA, local recurrence, ocular survival and metastasis between the two groups.¹⁸ More importantly, higher local recurrence rate was not observed when tumours were

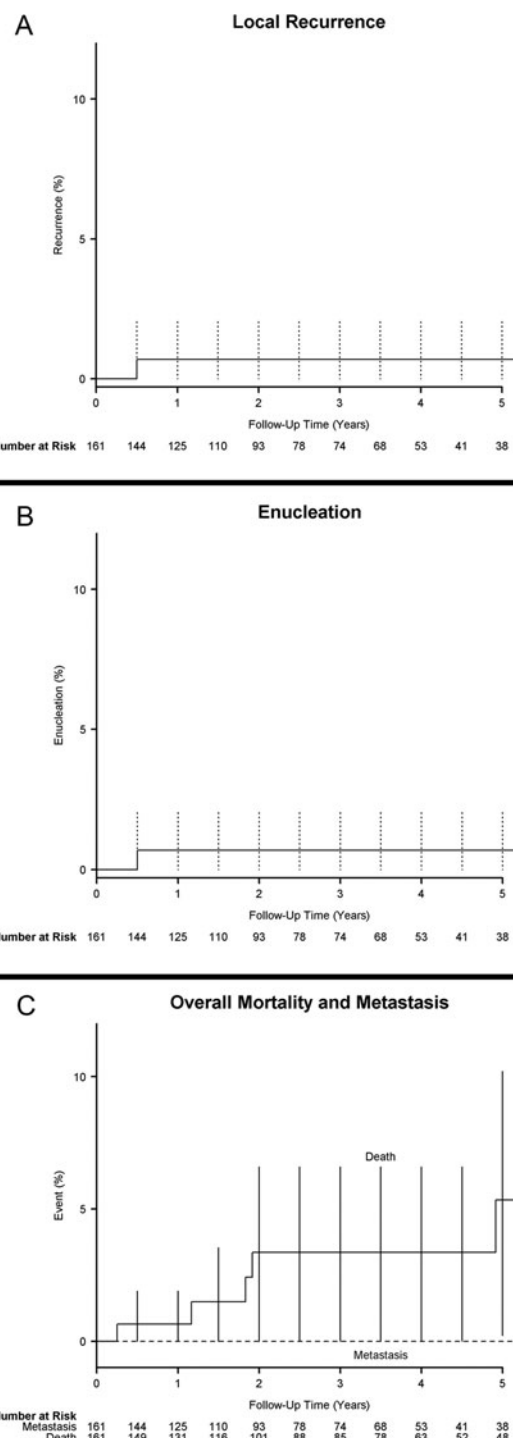


Figure 2 Kaplan-Meier estimation of local recurrence (99.3%) (A), ocular survival (99.3%) (B) and overall and metastasis-free survival (C) at 3 years following treatment. None of the study patients developed metastasis. Eight patients in this cohort died of unrelated causes. OS was 97% and MFS was 100% at 3 years. MFS, metastasis-free survival; OS, overall survival.

treated with apical height dose of 85 Gy (ABS guidelines).¹⁸ Taken together, the data show that prescription height dose of the COMS protocol^{1 14 15} can be considered supratherapeutic and is not expected to offer superior local control rates for small choroidal melanoma than those achieved by using ABS guidelines (99.3% at 3 years).

Table 3 Small melanoma outcomes: published studies: outcome measures

Author	Isotope	Outcome measures					
		Visual acuity		Recurrence	Ocular survival	Metastasis	Time (years)
		20/40–20/60	<20/200				
COMS ^{19*}	I-125	NR	NR	NR	NR	1%	5
Finger ²⁴	Pd-103	>20/50 (83%)	8%	0%	100%	0%	7
Damato ²⁰	Ru-106	NR	NR	3%	100%	NR	9
Isager ²¹	Ru-106	>20/60 (79%)	64%	0%	73%	NR	5
Verschueren ²²	Ru-106	60%	NR	2%	NR	NR	5
Sanchez-Tabernero ²⁵	I-125	NR	NR	0%	NR	NR	6.5 (Mean follow-up)
Shields ^{23*}	I-125, other†	NR	38%	7%	96%	1% (3 years) 3% (5 years)	3, 5
Present study	I-125	≥20/50 (69%)	13%	1%	99%	0%	3

*Calculated for tumours less than <2.5 mm (787 cases).

†Variety of radionuclides in addition to iodine-125. Also includes cases with prior therapy and additional transpupillary thermotherapy.

COMS, Collaborative Ocular Melanoma Study Report # 5; I, iodine; NR, not reported; Pd, palladium; Ru, ruthenium.

Only one event of local recurrence (that required enucleation) at 3 years following treatment was observed in the study, with a 3-year event-free rate of 99.3% for local recurrence and ocular survival (figure 2). Such high rates of local control have also been reported with alternate radionuclides used for brachytherapy such as Ru-106 (0%–3%)^{20–23} and Pd-103 (0%),²⁴ with dosing prescribed to tumour base rather than the apex,²² and even with other forms of radiation such as proton beam therapy (table 3).³¹ Consequently, ocular survival rates are also high (96%–100%).^{20 23 24} In a study of 375 eyes with posterior uveal melanoma (COMS: small and medium) that were treated with episcleral brachytherapy, local recurrence rate was 5.6%, the median time to recurrence was 18 months and the majority of recurrences (70%) occurred in the first 3 years.²⁸ It is certainly possible that local recurrence rates may be higher with longer follow-up.^{23 25} Although this was not observed in our study cohort, local recurrence seems to be an important risk factor associated with unusual cases of fatal small choroidal melanoma.³²

Metastasis was excluded at baseline by systemic staging (CT scan of chest abdomen and pelvis) followed by surveillance using hepatic ultrasonography every 6 months for the first 5 years and annually, thereafter.²⁷ None of the study patients developed metastasis while eight patients in this cohort died of unrelated causes. Our survival results are supported by the previous COMS report¹⁹ AJCC TNM cohort study³³ as well as two retrospective studies.^{19 23} However, no therapeutic benefit of intervention on survival can be inferred from this study as outcomes in patients with such tumours that were not treated are not known.

Since May 2009, we have offered prognostication to all patients with choroidal melanoma. Over the years, the techniques have evolved from fluorescent in situ hybridisation (FISH),³⁴ to multiplex ligation-dependent probe amplification (MLPA, Impact Genetics, Toronto, Canada). More recently, we have sent samples for commercially available gene expression profiling (GEP) test (DecisionDx-UM; Castle Biosciences, Phoenix, Arizona, USA).³⁵ Good prognosis patients were defined as those with tumours having disomy chromosome 3 as assessed by FISH/MLPA or Class 1 by GEP. Bad prognosis patients were defined as those with tumours having monosomy 3 by FISH/MLPA or Class 2 GEP. In this cohort of 161 patients, 23 underwent prognostication with the majority (21, 91%) having a good prognosis. Only two patients (9%) exhibited poor prognosis; yet over their

follow-up of 24 and 112 months, none of them developed metastasis. One of them had received adjuvant therapy, which was later deemed to be ineffective.³⁵ These observations are not in conflict with a general understanding that smaller tumours are more likely to have a good prognostic profile (approximately 95%)³⁶ and even if they had a poor prognostic profile, their survival outcome was better than larger tumours with a poor prognostic profile.^{35 37 38} In a major study of tumour size and prognostication, the only group of patients that did not develop metastasis were smaller tumours with a good prognostic profile.³⁹

The limitations of study include retrospective nature, sample size and follow-up duration. Bias inherent in any retrospective was minimised by using a standard protocol-based assessment for both ocular examination and systemic surveillance.^{27 28} Considering the rarity of uveal melanoma, the study sample is large. However, for assessment of low-frequency events such as recurrence and metastasis reported herein, larger number of samples (about 10-fold) would be desirable. The follow-up duration could be longer but seems to be adequate (mean 40.7 months, median 37.1 months) given that previous observations indicate that the median time to recurrence following brachytherapy is 18 months (range 4–156 months) with the majority (70%) of the recurrences occurring within the first 3 years.²⁸

Taken together, our data indicate that small choroidal melanoma treated with iodine-125 episcleral brachytherapy has excellent outcomes. The majority (69%) of patients retained VA of 20/50 or better with very high local control and ocular survival rate (99.3%) with the absence of metastasis at 3 years.

Contributors RJY: data collection, interpretation, drafting, critical revision of the manuscript. JB: data analysis and interpretation, critical revision of manuscript. AW: study design, data acquisition, data interpretation, drafting manuscript. JS: data interpretation, drafting of manuscript and interpretation. AS: study design, data interpretation, drafting manuscript, critical revision of the manuscript.

Funding The authors have not declared a specific grant for this research from any funding agency in the public, commercial or not-for-profit sectors.

Competing interests None declared.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement Data are available upon reasonable request.

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REFERENCES

- COMS Group. Design and methods of a clinical trial for a rare condition: the Collaborative Ocular Melanoma Study. COMS report no. 3. *Control Clin Trials* 1993;14:362–91.
- Diener-West M, JD E, SL F, *et al.* The COMS randomized trial of iodine 125 brachytherapy for choroidal melanoma, III: initial mortality findings. COMS report no. 18. *Arch Ophthalmol* 2001;119:969–82.
- COMS Group. Trends in size and treatment of recently diagnosed choroidal melanoma, 1987–1997: findings from patients examined at Collaborative Ocular Melanoma Study (COMS) centers: COMS report no. 20. *Arch Ophthalmol* 2003;121:1156–62.
- Aronow ME, Topham AK, Singh AD. Uveal melanoma: 5-year update on incidence, treatment, and survival (SEER 1973–2013). *Ocul Oncol Pathol* 2018;4:145–51.
- Shields JA. Treating some small melanocytic choroidal lesions without waiting for growth. *Arch Ophthalmol* 2006;124:1344–6.
- Murray TG, Sobrin L. The case for observational management of suspected small choroidal melanoma. *Arch Ophthalmol* 2006;124:1342–4.
- Augsburger JJ, Vrabec TR. Impact of delayed treatment in growing posterior uveal melanomas. *Arch Ophthalmol* 1993;111:1382–6.
- Kupfer C. Discussion: risk factors for growth and metastasis of small choroidal melanocytic lesions. *Trans Am Ophthalmol Soc* 1995;93:277–8.
- Shields JA, Shields CL. Management of posterior uveal melanoma: past, present, and future: the 2014 Charles L. Schepens lecture. *Ophthalmology* 2015;122:414–28.
- Singh AD. A small step towards big data: trends and preference patterns in ophthalmic oncology. *Ocular Oncol Pathol* 2015;1:39–53.
- Finger PT. Radiation therapy for choroidal melanoma. *Surv Ophthalmol* 1997;42:215–32.
- Seregard S, Pelayes DE, Singh AD. Radiation therapy: uveal tumors. *Dev Ophthalmol* 2013;52:36–57.
- Marwaha G, Macklis R, Singh AD, *et al.* Brachytherapy. *Dev Ophthalmol* 2013;52:29–35.
- Earle J, Kline RW, Robertson DM. Selection of iodine 125 for the Collaborative Ocular Melanoma Study. *Arch Ophthalmol* 1987;105:763–4.
- Melia BM, Abramson DH, Albert DM, *et al.* Collaborative Ocular Melanoma Study (COMS) randomized trial of I-125 brachytherapy for medium choroidal melanoma. I. Visual acuity after 3 years COMS report no. 16. *Ophthalmology* 2001;108:348–66.
- American Brachytherapy Society - Ophthalmic Oncology Task Force, Electronic Address pec, Committee AO. The American Brachytherapy Society consensus guidelines for plaque brachytherapy of uveal melanoma and retinoblastoma. *Brachytherapy* 2014;13:1–14.
- Echegaray JJ, Bechrakis NE, Singh N, *et al.* Iodine-125 brachytherapy for uveal melanoma: a systematic review of radiation dose. *Ocul Oncol Pathol* 2017;3:193–8.
- Murray TG, Markoe AM, Gold AS, *et al.* Long-term followup comparing two treatment dosing strategies of (125) I plaque radiotherapy in the management of small/medium posterior uveal melanoma. *J Ophthalmol* 2013;2013:517032.
- Group C. Mortality in patients with small choroidal melanoma. COMS report no. 4. The Collaborative Ocular Melanoma Study group. *Arch Ophthalmol* 1997;115:886–93.
- Damato B, Patel I, Campbell IR, *et al.* Local tumor control after 106Ru brachytherapy of choroidal melanoma. *Int J Radiat Oncol Biol Phys* 2005;63:385–91.
- Isager P, Ehlers N, Urbak SF, *et al.* Visual outcome, local tumour control, and eye preservation after 106Ru/Rh brachytherapy for choroidal melanoma. *Acta Oncol* 2006;45:285–93.
- Verschueren KM, Creutzberg CL, Schalijs-Delfos NE, *et al.* Long-term outcomes of eye-conserving treatment with ruthenium(106) brachytherapy for choroidal melanoma. *Radiother Oncol* 2010;95:332–8.
- Shields CL, Sioufi K, Srinivasan A, *et al.* Visual outcome and millimeter incremental risk of metastasis in 1780 patients with small choroidal melanoma managed by plaque radiotherapy. *JAMA Ophthalmol* 2018;136:1325–33.
- Finger PT, Berson A, Szechter A. Palladium-103 plaque radiotherapy for choroidal melanoma: results of a 7-year study. *Ophthalmology* 1999;106:606–13.
- Sanchez-Taberner S, Garcia-Alvarez C, Munoz-Moreno MF, *et al.* Pattern of local recurrence after i-125 episcleral brachytherapy for uveal melanoma in a spanish referral ocular oncology unit. *Am J Ophthalmol* 2017;180:39–45.
- Aziz HA, Al Zaharani YA, Bena J, *et al.* Episcleral brachytherapy of uveal melanoma: role of intraoperative echographic confirmation. *Br J Ophthalmol* 2017;101:747–51.
- Choudhary MM, Gupta A, Bena J, *et al.* Hepatic ultrasonography for surveillance in patients with uveal melanoma. *JAMA Ophthalmol* 2016;134:174–80.
- Bellerive C, Aziz HA, Bena J, *et al.* Local failure after episcleral brachytherapy for posterior uveal melanoma: patterns, risk factors, and management. *Am J Ophthalmol* 2017;177:9–16.
- Boldt HC, Murray TG, Binkley EM. Studies evaluating visual outcomes after brachytherapy in uveal melanoma: strengths and limitations of current investigations. *JAMA Ophthalmol* 2020;138:146–7.
- Aziz HA, Singh N, Bena J, *et al.* Vision loss following episcleral brachytherapy for uveal melanoma: development of a vision prognostication tool. *JAMA Ophthalmol* 2016;134:615–20.
- Lane AM, Egan KM, Kim IK, *et al.* Mortality after diagnosis of small melanocytic lesions of the choroid. *Arch Ophthalmol* 2010;128:996–1000.
- Jouhi S, Jager MJ, SJR DG, *et al.* The small fatal choroidal melanoma study. A survey by the European Ophthalmic Oncology Group. *Am J Ophthalmol* 2019;202:100–8.
- Kujala E, Damato B, Coupland SE, *et al.* Staging of ciliary body and choroidal melanomas based on anatomic extent. *J Clin Oncol* 2013;31:2825–31.
- Singh AD, Medina CA, Singh N, *et al.* Fine-needle aspiration biopsy of uveal melanoma: outcomes and complications. *Br J Ophthalmol* 2016;100:456–62.
- Binkley EM, Bena JF, Davanzo JM, *et al.* Gene expression profiling prognostication of posterior uveal melanoma: does size matter? *Ophthalmol Retina* 2020;4:620–9.
- Berry D, Seider M, Stinnett S, *et al.* relationship of clinical features and baseline tumor size with gene expression profile status in uveal melanoma: a multi-institutional study. *Retina* 2019;39:1154–64.
- Correa ZM, Augsburger JJ. Independent prognostic significance of gene expression profile class and largest basal diameter of posterior uveal melanomas. *Am J Ophthalmol* 2016;162:e1.
- Walter SD, Chao DL, Feuer W, *et al.* Prognostic implications of tumor diameter in association with gene expression profile for uveal melanoma. *JAMA Ophthalmol* 2016;134:734–40.
- Dogrusoz M, Bagger M, van Duinen SG, *et al.* The prognostic value of AJCC staging in uveal melanoma is enhanced by adding chromosome 3 and 8q status. *Invest Ophthalmol Vis Sci* 2017;58:833–42.